

DATA SCIENCE CAPSTONE

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EXECUTIVE SUMMARY

Methodologies Used in this Analysis:

- Data Collection
- Data Wrangling
- Exploratory Data Analysis with Data Visualization
- Exploratory Data Analysis with SQL
- Interactive Maps Using Folium
- Interactive Dashboards Using Plotly Dash
- Predictive Analysis

Results:

- EDA Results
- Interactive Maps and Dashboards via Images
- Machine Learning Models Results

INTRODUCTION

SpaceX revolutionized space travel with its affordable launch costs thanks to reusable Falcon 9 rockets. Predicting the reusability of the first stage helps estimate launch costs. By leveraging machine learning and publicly available data, this project aims to build a model that predicts successful first-stage landings.

In this analysis we aimed to answer three specific questions:

1. How do certain variables such as where the rocket was launched, or number of flights and orbits affect the success of the first stage landing?
2. Have the success of the first stage successfully landing improved over time?
3. What is the most accurate machine learning model that can be used to predict the first stage landing successfully?

METHODOLOGY



DATA COLLECTION

Data was collected from two sources using two retrieval processes.

1. SpaceX REST API
2. Web scraping Wikipedia via BeautifulSoup



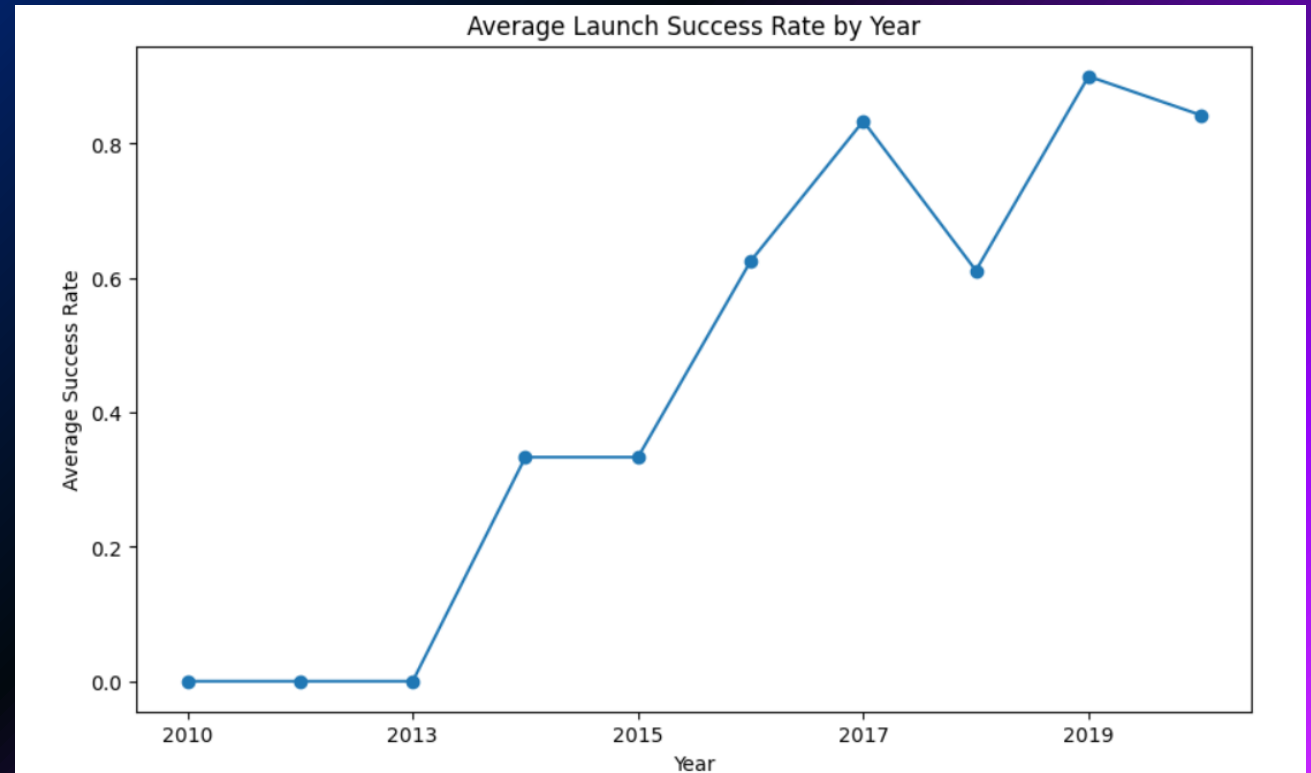
DATA WRANGLING

Data was analyzed for missing and incorrect values. These missing values were then updated with the mean value for their respective variable.

Feature engineering was performed to translate qualitative values to quantitative values that can be used in models.

EXPLORATORY DATA ANALYSIS

Visualized and analyzed the data using both traditional Python methods and SQL. This allowed us to find relationships and trends in the data that could then be applied in the data modeling stage.



EXPLORATORY DATA ANALYSIS WITH SQL

SQL Queries:

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'KSC'A
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the date where the successful landing outcome in drone ship was achieved.
- List the names of the boosters which have success in ground pad and have payload mass greater than 4000 but less than 6000
- List the total number of successful and failure mission outcomes
- List the names of the booster versions which have carried the maximum payload mass. Use a subquery
- List the records which will display the month names, successful landing outcomes in ground pad ,booster versions, launch site for the months in year 2017
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

EXPLORATORY DATA ANALYSIS WITH VISUALIZATION

Chart Types Used:

- Scatter Plots
 - Show relationship between two variables
- Box Charts
 - Comparison among categorical variables
- Line Plots
 - Show trends over time

Charts:

- Flight Number & Launch Site
- Payload Mass & Launch Site
- Success Rate of Each Orbit Type
- Flight Number & Orbit Type
- Payload Mass & Orbit Type
- Yearly Launch Success Trend

Feature Engineering:

- Adding in Dummy Variables

INTERACTIVE MAP WITH FOLIUM

Interactive Exploration

•Launch Site Selection:

- Interactive dropdown list for selecting a specific Launch Site.

•Launch Success Analysis:

- Pie Chart:
 - Overall successful launches for all sites.
 - Success vs. Failed launches for the selected Launch Site (if applicable).

•Payload Mass Analysis:

- Slider: Adjusts the payload mass range for data filtering.
- Scatter Chart:
 - Visualizes the correlation between Payload Mass and Launch Success for different Booster Versions.

DASHBOARD WITH PLOTLY DASH

Launch Site Visualization

•All Launch Sites:

- Displayed using Markers with Circle, Popup Labels, and Text Labels.
- Locations pinpointed by latitude and longitude coordinates.
- Visualizes geographical distribution, proximity to the Equator, and coastal locations.

•Launch Outcome:

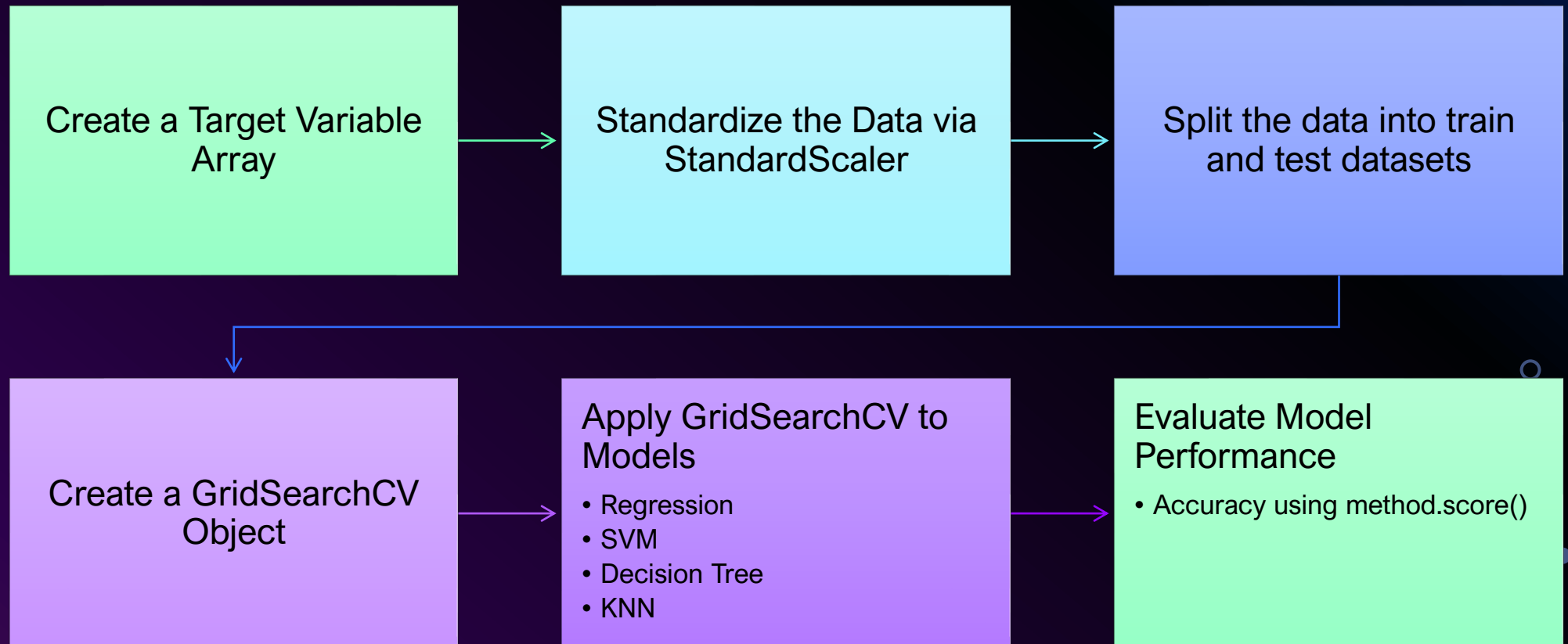
- Color-coded Markers: Green for successful launches, Red for failed launches.
- Utilizes Marker Clustering to identify launch sites with high success rates.

•Proximity Analysis:

- Demonstrates distances between a selected Launch Site (e.g., KSC LC-39A) and key proximities:
 - Railway
 - Highway
 - Coastline
 - Closest City
- Visualized using colored Lines.

[GitHub URL](#)

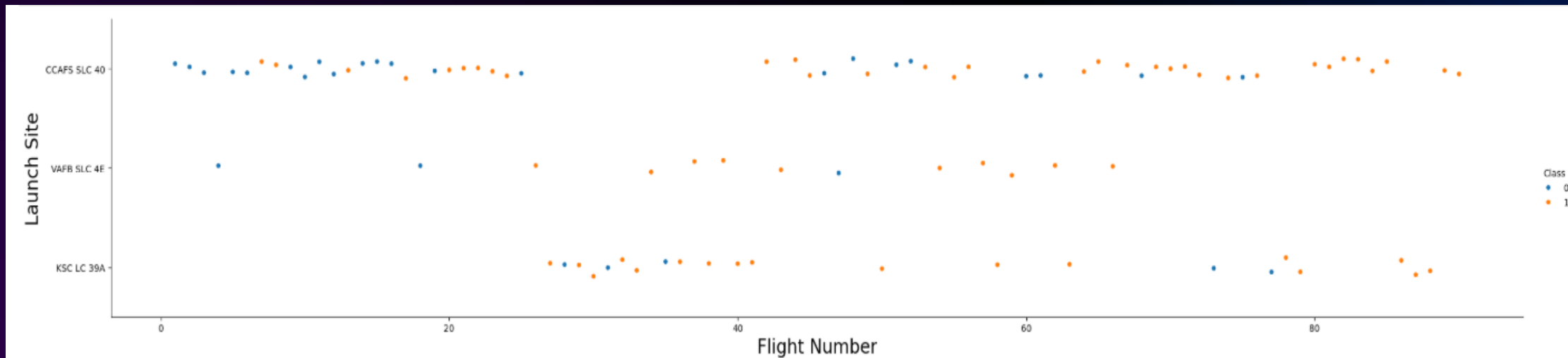
PREDICTIVE ANALYSIS (CLASSIFICATION) PROCESS



RESULTS

EDA WITH VISUALIZATION

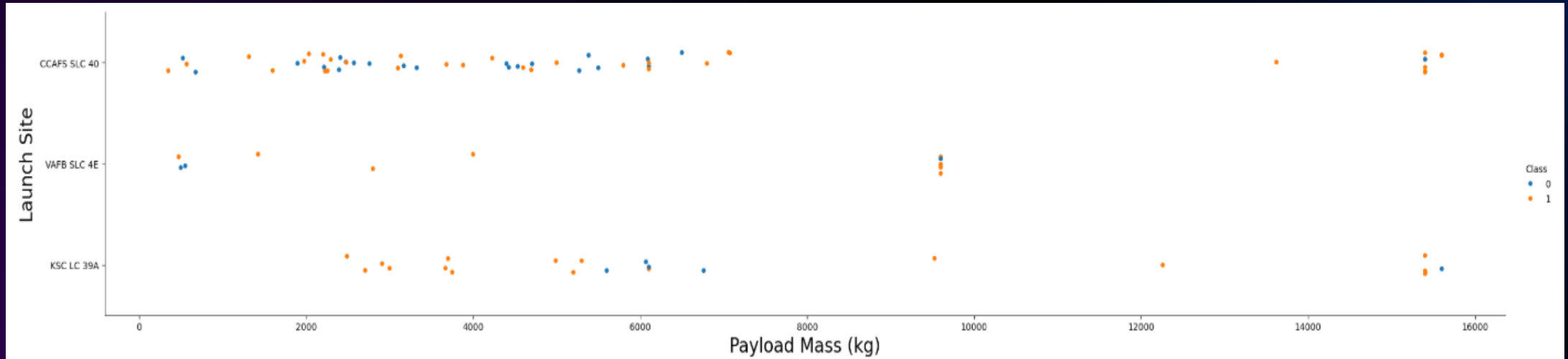
FLIGHT NUMBER & LAUNCH SITE



Findings:

- Launch success got better over time. Majority of launches failed early on.
- Site CCAFS SLC 40 has the most launches
- Sites VAFB SLC 4E & KSC LC 39A have higher success rates even though they have fewer total launches than CCAFS SLC 40.

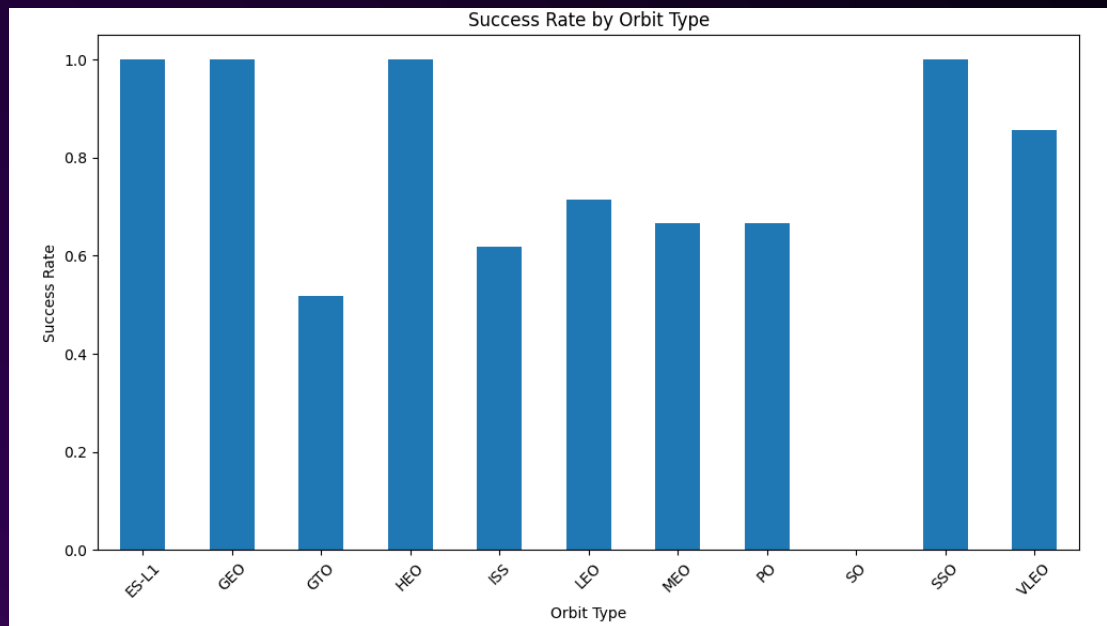
PAYLOAD MASS & LAUNCH SITE



Findings:

- Almost all launches with over 7,000 kgs was successful
- Almost all launches under 6,000 kgs at site KSC LC 39A was successful

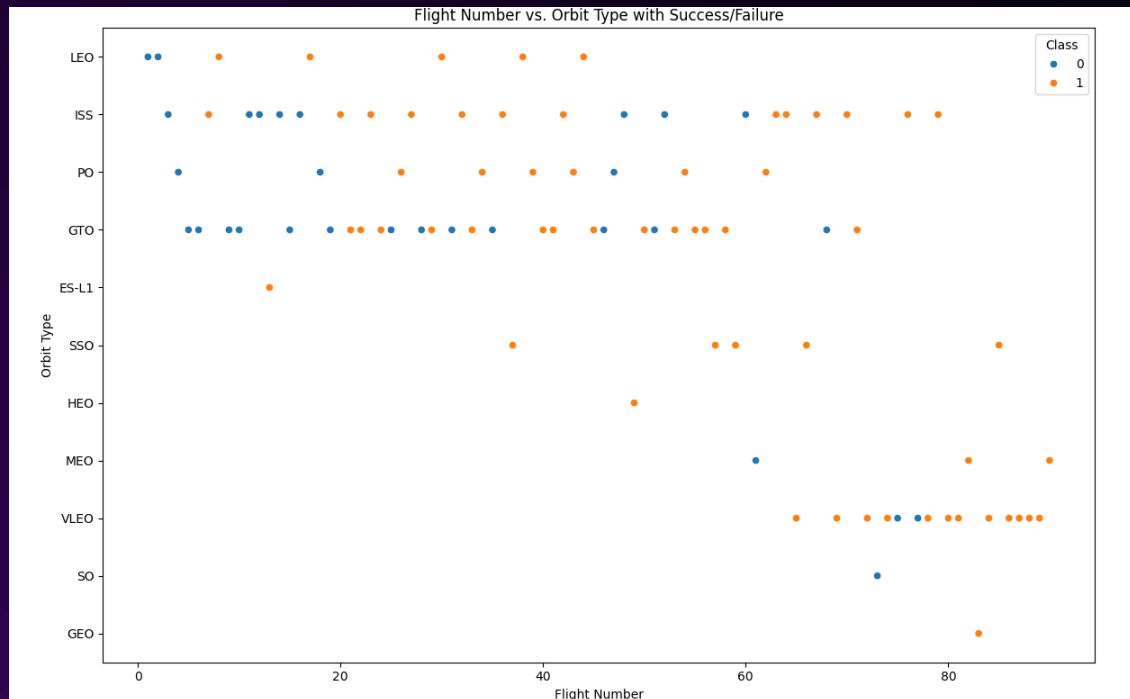
SUCCESS RATE FOR ORBIT TYPE



Findings:

- 4 orbit types have 100% success rates
- 5 orbit types were below ~65% success rate
- Need to know how many launch attempts per Orbit Type
- Only launch during those 4 orbit types if possible

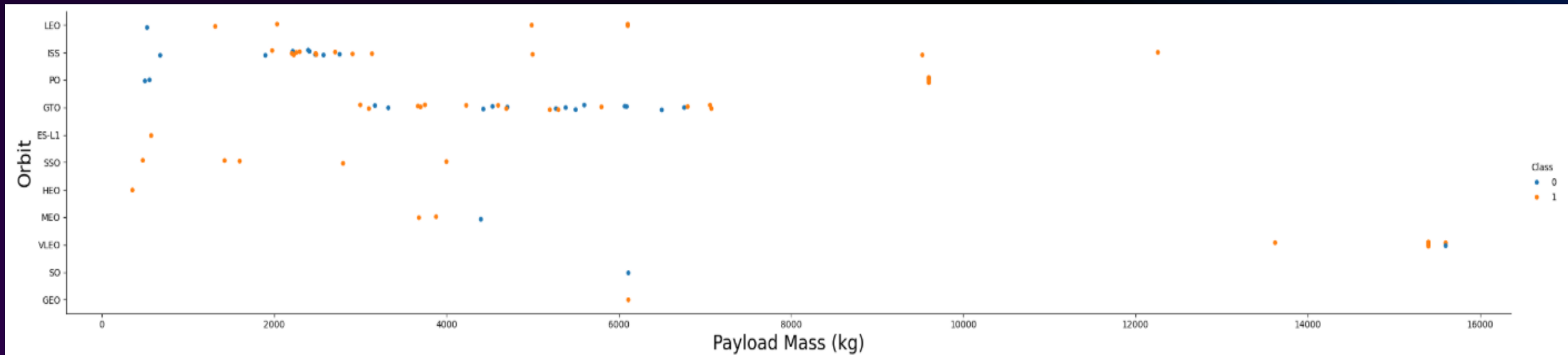
FLIGHT NUMBER & ORBIT TYPE



Findings:

- Orbit types with 100% success have minimal launches
- Type GTO has many launches and a low success rate
- VLEO is a newer orbit type but has a high success rate

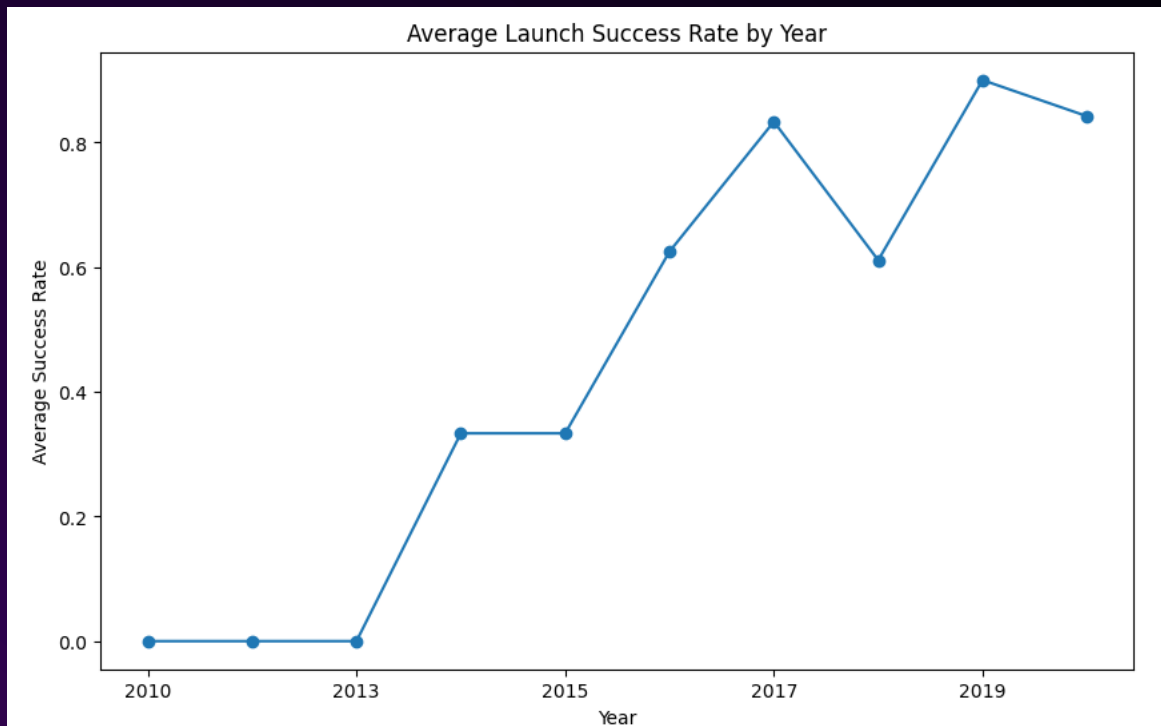
PAYLOAD MASS & ORBIT TYPE



Findings:

- Almost all launches with over 8,000 kgs was successful for all orbit types

SUCCESS RATE BY YEAR



Findings:

- Launch success improved or stayed the same every year except for 2018 and 2020
- Would want to also see total launches by year to ensure those two years had a valid number of launches. COVID could have impacted launches in 2020.

EDA WITH SQL

DISPLAY THE NAMES OF THE UNIQUE LAUNCH SITES IN THE SPACE MISSION

```
[8]: %sql SELECT DISTINCT "Launch_Site" from SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[8]: Launch_Site
```

```
CCAFS LC-40
```

```
VAFB SLC-4E
```

```
KSC LC-39A
```

```
CCAFS SLC-40
```

DISPLAY 5 RECORDS WHERE LAUNCH SITES BEGIN WITH THE STRING 'KSC'

```
[10]: %sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE ('KSC%') LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

```
[10]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
2017-03-16	6:00:00	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar	Success	No attempt
2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
2017-05-01	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	Success	Success (ground pad)
2017-05-15	23:21:00	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat	Success	No attempt

DISPLAY THE TOTAL PAYLOAD MASS CARRIED BY BOOSTERS LAUNCHED BY NASA (CRS)

```
[16]: %sql SELECT SUM(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)' GROUP BY Customer;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[16]: SUM(PAYLOAD_MASS_KG_)
```

```
45596
```

DISPLAY AVERAGE PAYLOAD MASS CARRIED BY BOOSTER VERSION F9 V1.1

```
[17]: %sql SELECT AVG(PAYLOAD_MASS_KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1' GROUP BY Booster_Version;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[17]: AVG(PAYLOAD_MASS_KG_)
```

```
2928.4
```

LIST THE DATE WHERE THE SUCCESSFUL LANDING OUTCOME IN DRONE SHIP WAS ACHIEVED

```
[18]: %sql SELECT min(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)';
      * sqlite:///my_data1.db
      Done.
[18]: min(Date)
      2016-04-08
```

LIST THE NAMES OF THE BOOSTERS WHICH HAVE SUCCESS IN GROUND PAD AND HAVE PAYLOAD MASS GREATER THAN 4000 BUT LESS THAN 6000

```
[24]: %sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)' AND PAYLOAD_MASS_KG_ BETWEEN 4000 AND 6000 GROUP BY Booster_Version;
* sqlite:///my_data1.db
Done.
[24]: Booster_Version
```

F9 B4 B1040.1
F9 B4 B1043.1
F9 FT B1032.1

LIST THE TOTAL NUMBER OF SUCCESSFUL AND FAILURE MISSION OUTCOMES

```
[26]: %sql SELECT COUNT(*), Mission_Outcome FROM SPACEXTABLE GROUP BY Mission_Outcome;
```

```
* sqlite:///my_data1.db
```

```
Done.
```

```
[26]:
```

COUNT(*)	Mission_Outcome
1	Failure (in flight)
98	Success
1	Success
1	Success (payload status unclear)

LIST THE NAMES OF THE BOOSTER VERSIONS WHICH HAVE CARRIED THE MAXIMUM PAYLOAD MASS. USE A SUBQUERY

```
[9]: %sql SELECT Booster_Version FROM SPACEXTABLE WHERE payload_mass__kg_ = (select max(PAYLOAD_MASS__KG_) FROM SPACEXTABLE);
```

```
* sqlite:///my_data1.db  
Done.
```

```
[9]: Booster_Version
```

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

LIST THE RECORDS WHICH WILL DISPLAY THE MONTH NAMES, SUCCESSFUL LANDING OUTCOMES IN GROUND PAD ,BOOSTER VERSIONS, LAUNCH SITE FOR THE MONTHS IN YEAR 2017

```
[17]: %%sql SELECT substr(Date,6,2) AS month, substr(Date,9,2) AS day, Booster_Version, Launch_Site, Landing_Outcome from SPACEXTABLE
      where Landing_Outcome = 'Success (ground pad)' and substr(Date,0,5)='2017';
```

```
* sqlite:///my_data1.db
```

Done.

```
[17]:
```

month	day	Booster_Version	Launch_Site	Landing_Outcome
02	19	F9 FT B1031.1	KSC LC-39A	Success (ground pad)
05	01	F9 FT B1032.1	KSC LC-39A	Success (ground pad)
06	03	F9 FT B1035.1	KSC LC-39A	Success (ground pad)
08	14	F9 B4 B1039.1	KSC LC-39A	Success (ground pad)
09	07	F9 B4 B1040.1	KSC LC-39A	Success (ground pad)
12	15	F9 FT B1035.2	CCAFS SLC-40	Success (ground pad)

RANK THE COUNT OF LANDING OUTCOMES (SUCH AS FAILURE (DRONE SHIP) OR SUCCESS (GROUND PAD)) BETWEEN THE DATE 2010-06-04 AND 2017-03-20, IN DESCENDING ORDER

```
[19]: %%sql SELECT Landing_Outcome, count(*) as count_outcomes
      FROM SPACEXTABLE
      WHERE Date between '2010-06-04' and '2017-03-20'
      GROUP BY Landing_Outcome
      ORDER BY count_outcomes desc;
```

```
* sqlite:///my_data1.db
Done.
```

```
[19]:
```

Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

INTERACTIVE MAP WITH FOLIUM

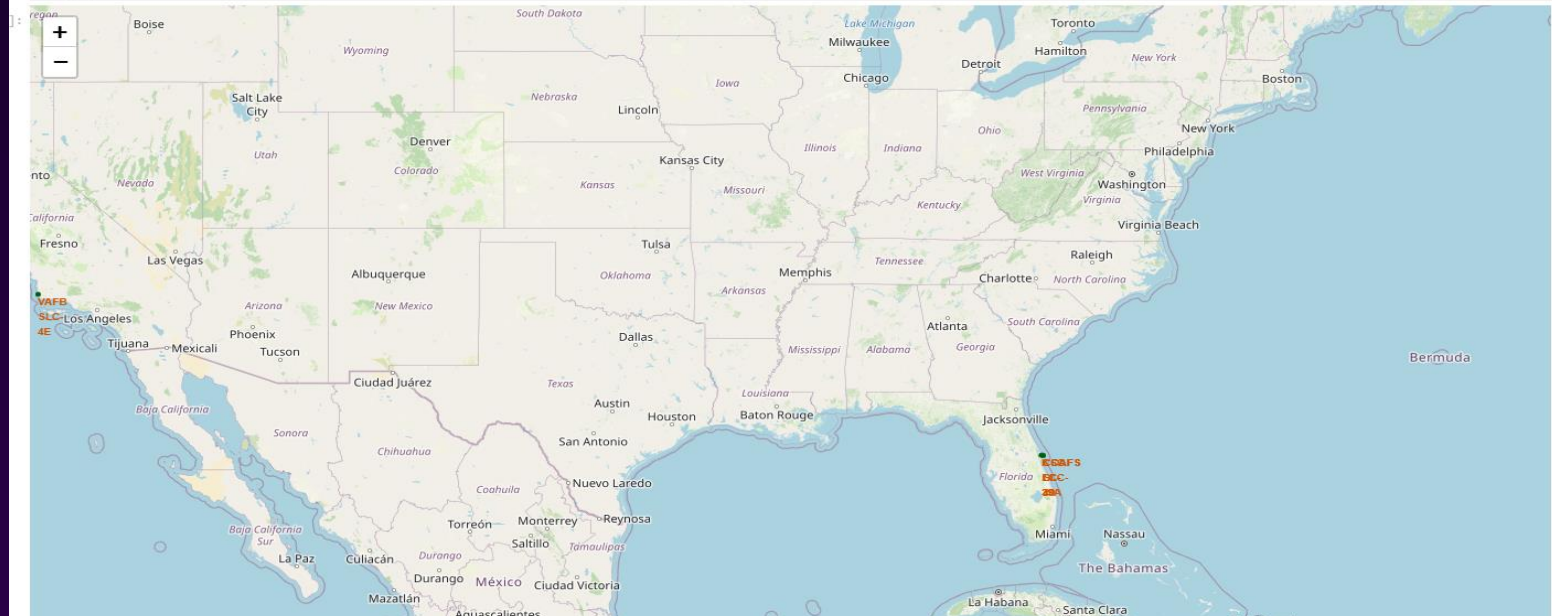
FOLIUM MAP WITH LABELS

```
for index, row in launch_sites_df.iterrows():
    coordinate = [row['Lat'], row['Long']]

    circle = folium.Circle(coordinate, radius=1000, color="#006008", fill=True)
    circle.add_child(folium.Popup(row['Launch Site'])) # Assuming 'LaunchSiteName' is the column name
    site_map.add_child(circle)

    marker = folium.Marker(coordinate, icon=folium.DivIcon(icon_size=(20, 20), icon_anchor=(0, 0),
                                                            html='<div style="font-size: 12; color: #d35400;"><b>{}</b></div>'.format(row['Launch Site'])))
    site_map.add_child(marker)
```

site_map



FOLIUM MAP COLOR FOR SUCCESS FAILURE

```
[15]: # Add marker_cluster to current site_map
site_map.add_child(marker_cluster)

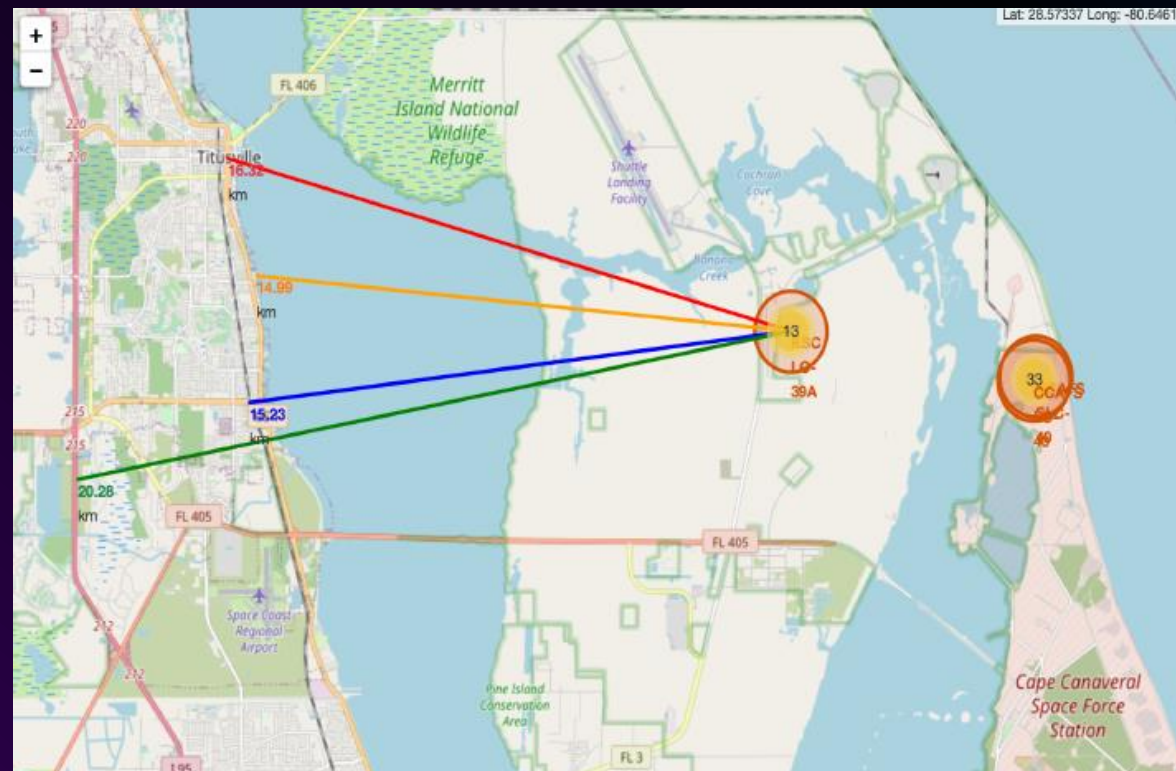
# for each row in spacex_df data frame
# create a Marker object with its coordinate
# and customize the Marker's icon property to indicate if this launch was succeeded or failed,
# e.g., icon=folium.Icon(color='white', icon_color=row['marker_color'])
# Add markers to the cluster
for index, row in spacex_df.iterrows():
    marker = folium.Marker(location=[row['Lat'], row['Long']],
                           icon=folium.Icon(color=row['marker_color'], icon="fa-rocket"))
    marker_cluster.add_child(marker)

# Create the map and add the marker cluster
site_map = folium.Map(location=[launch_sites_df['Lat'].mean(), launch_sites_df['Long'].mean()], zoom_start=5)
site_map.add_child(marker_cluster)

# Display the map
site_map
```



FOLIUM MAP WITH DISTANCE MARKERS



DASHBOARD WITH PLOTLY DASH

COMPLETED DASHBOARD USING PLOTLY DASH

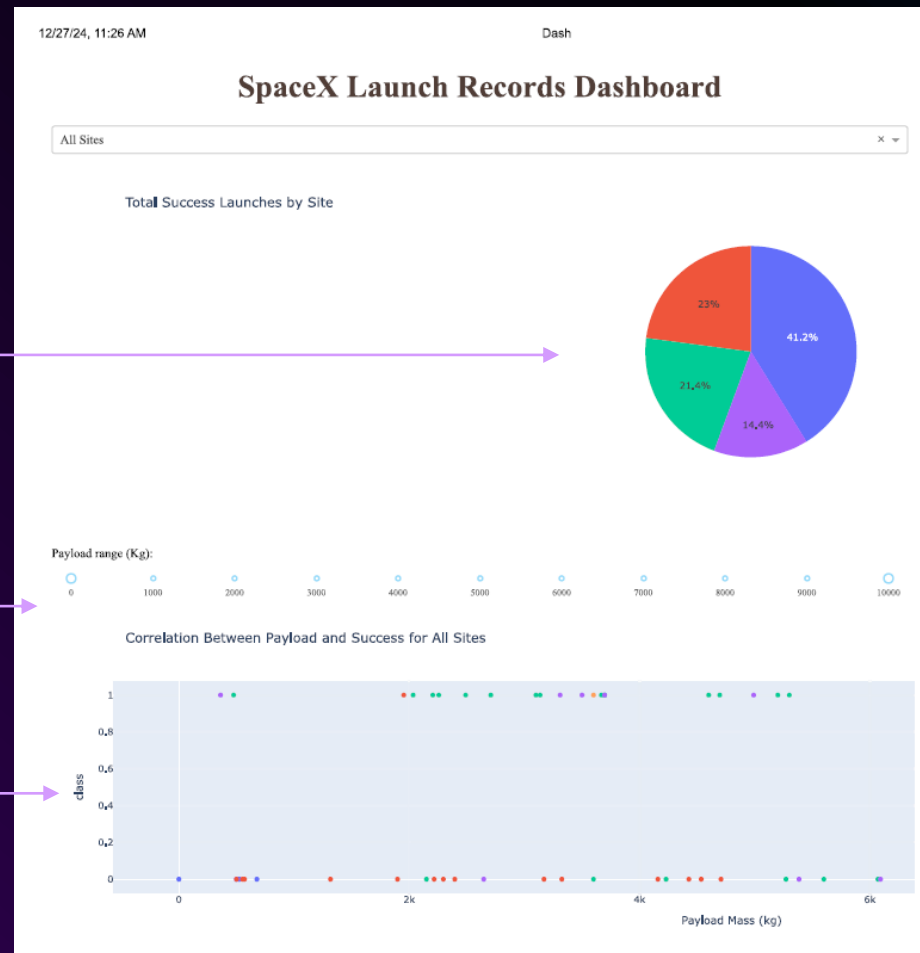
Site Filter

Success Pie Chart

Payload Slicer

Payload & Success Rate

[GitHub URL](#)



Findings:

- KSC LC 39A has highest success rate
- Payload between 2,000 and 5,500 kgs have the highest success rate

PREDICTIVE MODELING

CLASSIFICATION ACCURACY & FINAL FINDINGS

Testing Data

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.800000	0.800000	0.923077	0.800000
F1_Score	0.888889	0.888889	0.960000	0.888889
Accuracy	0.833333	0.833333	0.944444	0.833333

Entire Dataset

	LogReg	SVM	Tree	KNN
Jaccard_Score	0.833333	0.845070	0.857143	0.819444
F1_Score	0.909091	0.916031	0.923077	0.900763
Accuracy	0.866667	0.877778	0.900000	0.855556

Findings:

- Decision Tree has the highest accuracy scores for predicting whether stage 1 will land successfully or not.
- All other models performed similarly well

THANK YOU

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